

Cognizant Digital Nurture 5.0

Java Full Stack Engineer (Angular Track)

Week 2 Learning Progress Report

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Program: Cognizant Digital Nurture 5.0

Track: Java Full Stack Engineer (Angular)

Week: Week 2

Topics Covered

Spring Core and Maven

Spring Data JPA with Spring Boot

Hibernate

Objective

The objective of Week 2 is to understand the fundamentals of the Spring Framework and Maven. This includes learning how Spring manages objects using Inversion of Control (IoC) and Dependency Injection (DI), configuring beans, building projects with Maven, and developing enterprise applications using Spring Boot. These concepts provide the foundation for developing scalable and maintainable Java applications.

Prepared By

Pooja V

Cognizant Digital Nurture 5.0 – Deep Skilling Program

TDD using JUnit5 and Mockito

File 1: Calculator.java

```
public class Calculator {  
  
    public int add(int a, int b) {  
        return a + b;  
    }  
}
```

File 2: CalculatorTest.java

```
import org.junit.jupiter.api.Test;  
import static org.junit.jupiter.api.Assertions.assertEquals;  
  
public class CalculatorTest {  
  
    @Test  
    public void testAddition() {  
  
        Calculator calculator = new Calculator();  
  
        int result = calculator.add(10, 20);  
  
        assertEquals(30, result);  
    }  
}
```



The screenshot shows an IDE console window with the following error message:

```
<terminated> CalculatorTest [JUnit] C:\Users\user\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64.21.0.10.v20260205-0638\jre\bin\javaw.exe (29-Jun-2026, 2:58:44 pm - 2:58:44 pm)  
Caused by: java.lang.NoClassDefFoundError: org/junit/platform/engine/ConfigurationParameters  
    at org.eclipse.jdt.internal.junit5.runner.JUnit5TestLoader.<init>(JUnit5TestLoader.java:37)  
    at java.base/jdk.internal.reflect.DirectConstructorHandleAccessor.newInstance(DirectConstructorHandleAccessor.java:62)  
    ... 7 more  
Caused by: java.lang.ClassNotFoundException: org.junit.platform.engine.ConfigurationParameters  
    at java.base/jdk.internal.loader.BuiltinClassLoader.loadClass(BuiltinClassLoader.java:641)  
    at java.base/jdk.internal.loader.ClassLoaders$AppClassLoader.loadClass(ClassLoaders.java:188)  
    at java.base/java.lang.ClassLoader.loadClass(ClassLoader.java:526)  
    ... 9 more  
  
at org.eclipse.jdt.internal.junit.runner.RemoteTestRunner.createRawTestLoader(RemoteTestRunner.java:377)  
at org.eclipse.jdt.internal.junit.runner.RemoteTestRunner.createLoader(RemoteTestRunner.java:367)  
at org.eclipse.jdt.internal.junit.runner.RemoteTestRunner.defaultInit(RemoteTestRunner.java:311)  
at org.eclipse.jdt.internal.junit.runner.RemoteTestRunner.init(RemoteTestRunner.java:226)  
at org.eclipse.jdt.internal.junit.runner.RemoteTestRunner.main(RemoteTestRunner.java:210)
```

File 1: Calculator.java

```
package junitexercise;

public class Calculator {

    public int add(int a, int b) {
        return a + b;
    }

    public int subtract(int a, int b) {
        return a - b;
    }

    public int multiply(int a, int b) {
        return a * b;
    }
}
```

File 2: CalculatorAAATest.java

```
package junitexercise;

import static org.junit.jupiter.api.Assertions.assertEquals;

import org.junit.jupiter.api.AfterEach;
import org.junit.jupiter.api.BeforeEach;
import org.junit.jupiter.api.Test;

public class CalculatorAAATest {

    private Calculator calculator;

    @BeforeEach
    public void setUp() {
        System.out.println("Setting up Calculator...");
        calculator = new Calculator();
    }

    @AfterEach
    public void tearDown() {
        System.out.println("Cleaning up...");
        calculator = null;
    }
}
```

```

@Test
public void testAddition() {

    // Arrange
    int a = 10;
    int b = 20;

    // Act
    int result = calculator.add(a, b);

    // Assert
    assertEquals(30, result);
}
}

```



Configuring a Basic Spring Application

HelloWorld.java

```
package com.example;

public class HelloWorld {

    public void display() {
        System.out.println("Welcome to Spring Framework!");
    }
}
```

App.java

```
package com.example;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class App {

    public static void main(String[] args) {

        ApplicationContext context =
            new ClassPathXmlApplicationContext("applicationContext.xml");

        HelloWorld hello = (HelloWorld) context.getBean("helloBean");

        hello.display();
    }
}
```

applicationContext.xml

```
<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

    xsi:schemaLocation="
```

```
http://www.springframework.org/schema/beans  
https://www.springframework.org/schema/beans/spring-beans.xsd">
```

```
<bean id="helloBean"  
      class="com.example.HelloWorld"/>
```

```
</beans>
```

Implementing Dependency Injection

Engine.java

```
package com.example;

public class Engine {

    public void start() {
        System.out.println("Engine Started");
    }
}
```

Car.java

```
package com.example;

public class Car {

    private Engine engine;

    public Car(Engine engine) {
        this.engine = engine;
    }

    public void drive() {

        engine.start();

        System.out.println("Car is Running");
    }
}
```

App.java

```
package com.example;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class App {
```

```
public static void main(String[] args) {  
  
    ApplicationContext context =  
        new ClassPathXmlApplicationContext("applicationContext.xml");  
  
    Car car = context.getBean("car", Car.class);  
  
    car.drive();  
}  
}
```

applicationContext.xml

```
<?xml version="1.0" encoding="UTF-8"?>  
  
<beans xmlns="http://www.springframework.org/schema/beans"  
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"  
  
    xsi:schemaLocation="  
        http://www.springframework.org/schema/beans  
        https://www.springframework.org/schema/beans/spring-beans.xsd">  
  
    <bean id="engine"  
        class="com.example.Engine"/>  
  
    <bean id="car"  
        class="com.example.Car">  
  
        <constructor-arg ref="engine"/>  
  
    </bean>  
  
</beans>
```